

Geometry Midterm

January 27, 2025

Problem 1

Restate the following in statements in an "if α then β " form so that α implies β .

- a) I will pay you \$20 only if you mow the lawn.
- b) Everyone who passed the test got an A in the class.
- c) Only the brave are free.

Problem 2

I will give you an A only if you pass this test.

- a) What can you conclude if I give you an A?
- b) What can you conclude if you pass the test?

Problem 3

If P and Q are points in the interior of $\angle O$, prove that the segment PQ is in the interior of $\angle O$. (Hint: proof by contradiction)

Problem 4

Prove that every line has a point on either side of it.

Problem 5

In $\angle AOB$, where $\angle AOB$ is NOT a straight angle, point P lies between A and B . Prove that P is in the interior of the angle.

Problem 6

Prove that if $AB \not\cong CD$ then $AB \neq CD$.

Problem 7

Prove that $0.\overline{9} = 1$.

Problem 8

A line l does not contain any of the vertices of a triangle $\triangle ABC$. The line l intersects side AB , but does NOT intersect side BC . Prove that l must intersect the side AC .

Problem 9

Prove that the base angles of an isosceles triangle are congruent.

Problem 10

Prove that any two right angles are congruent.

Problem 11

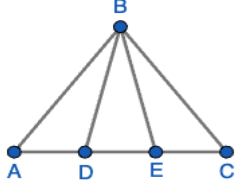
Prove that the angles opposite the legs of a right triangle are acute angles.

Problem 12

Prove that there is at most one line perpendicular to a given line l from a given point P . [Hint: Suppose that there were two perpendiculars, and derive a contradiction]

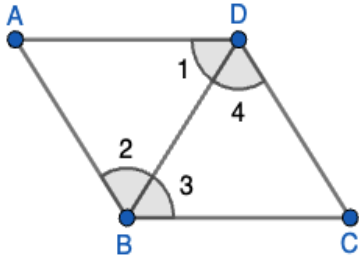
Problem 13

If $\triangle DBE$ is isosceles with $DB \cong EB$ and $\angle ABD \cong \angle CBE$, prove that $\triangle ABC$ is isosceles.



Problem 14

If $\angle 2 \cong \angle 4$ and $AD \cong BC$, prove that $\angle 1 \cong \angle 3$.



Problem 15

Determine if each of the following is true or false.

- a) If a theorem is true, its contrapositive is true.
- b) The supplement of a 40° angle is a 50° angle.
- c) If segment AB intersects segment CD at point O , then $\angle AOB \cong \angle BOD$.

Extra Credit

Prove that the angle sum of any triangle is a straight angle.